Kolikipogu Manas Karthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.14
Narayana JR College	Class XII (TELANGANA BOARD OF INTERMEDIATE EDUCATION)	2024	96.30%
NARAYANA OLYMPIAD HIGH SCHOOL	Class X (Board of Secondary Education Telangana)	2022	10.00%

PROJECTS

Traffic Light Controller (Verilog)

- Designed a traffic light controller in Verilog using a Finite State Machine (FSM) for traffic signal sequencing and timing control.
- Developed as the final project for the **Digital Systems (EE1201)** course.
- GitHub: ManasKarthik/EE1501-FP: Digital Systems Lab final project
- Collaborated with teammates to design the FSM state diagram, implement TrafficFSM.v, and verify functionality using simulation waveforms and an optimized TrafficLights_tb.v testbench.

Mod-100 Synchronous up/down counter (Breadboard)

- Built a two-digit synchronous BCD up/down counter on a breadboard using digital logic ICs.
- Demonstrated counting functionality and direction control.
- Developed as the final project for the **Electrical Circuits Lab (EE1200)** course.
- Debugged hardware issues caused by loose breadboard connections, faulty ICs, and edge-case behavior during testing.
- GitHub: ManasKarthik/Mod-100-Synchronous-up-down-counter: Final Project as part of course Electrical Circuits Lab (EE-1200)

RF Power Meter (PCB Design EE4130)

- Designed the PCB layout for an RF Power Meter and is currently debugging and validating the hardware.
- Routed a two-layer RF PCB with controlled analog signal paths, power distribution, ESP32 peripherals, USB-C charging, ADS1115 ADC, and OLED interface while addressing placement and routing constraints.
- Addressed late-stage design modifications and gained practical experience with PCB fabrication and manufacturing workflows.
- GitHub: ManasKarthik/RF-Power-Meter: Semester project for EE4130 containing design files and Altium project for RF Power Meter

Battery Management System (ESDP Lab EE2504)

- Developed a Battery Management System for an 18-cell battery stack on a four-layer PCB
- Contributed to PCB trace routing, ESP32 peripheral design, PCB fabrication preparation, and hardware debugging.
- Gained practical experience in high-voltage PCB design, handling fine-pitch electronic components, and identifying and troubleshooting hardware faults.
- GitHub: ManasKarthik/BMS-system-V1: semester project for EE2504 containing design files and Altium project for wireless board for ADBMS1818 AFE

N-bit Parameterised ALU

- Designed a parameterised Arithmetic Logic Unit (ALU) in Verilog using carry-lookahead and ripple-carry adders for addition and subtraction.
- Planned future extensions to support signed arithmetic, division, and floating-point operations.
- Developed as a personal project to explore different digital design architectures and improve Verilog design skills.

- GitHub: ManasKarthik/N-bit-Parameterized-ALU: Developed as a personal project to explore different digital design architectures and improve Verilog design skills.

SKILLS

Languages: C, Verilog

Hardware Design: Altium Designer, KiCad, LTspice

Development Tools: Vivado, Git, Linux, Vim

Concepts: Digital Logic Design, FSM Design, Computer Architecture

Documentation: LaTeX, Microsoft Word

RELEVANT COURSES

Theory:

Circuits and Network Analysis, Microprocessor and Computer Architecture, Signals and Systems, **Digital Systems**, Probability and Random Processes, Matrix Theory, Semiconductor Device Fundamentals, Numerical Methods for Device Modelling, Productivity Tools for IC design, Analog Electronics, Introduction to CMOS Processing, Control Theory for Circuits, Engineering Electromagnetics, **Introduction to Programming, Introduction to HDL**.

Lab:

Digital Systems Lab, Semiconductor Devices Lab, **Electronic Devices and Circuits Lab**, Analog Circuits Lab, **Electronic System Design Project Lab, PCB Design and Layout for electronic systems**.